

Elliot Beck

+41 79 214 00 25
elliott.beck@hotmail.com
github.com/elliottbeck
linkedin.com/in/elliott-beck

Relocating to London — UK HPI visa, no sponsorship needed

PROFILE

Macro researcher with six years at the Swiss National Bank building the pipelines, models and views behind the Bank's inflation and GDP forecasts: scraping news and alternative data, turning it into signals that run live in production, building option-implied distributions around policy events, and defending the resulting view to people who have to act on it. Every forecast is evaluated out of sample in real time. PhD in Finance (summa cum laude, UZH; visiting at Chicago Booth), MSc Statistics (ETH Zurich), four papers on forecasting, factor-model inference and news-based macro signals.

WORK EXPERIENCE

Swiss National Bank

Zurich, Switzerland

Data Scientist / Economist, Economic Data Science

Mar 2020 – present

- Built a news-based real-activity signal for Switzerland end to end: automated sourcing of Swiss media, LLM-based extraction of an economic-outlook index (LangChain, PyTorch), integration into the live GDP nowcast. Unstructured text now moves a published number
- Rebuilt the 1–12-month-ahead inflation forecasting pipeline around machine and deep learning methods (scikit-learn, PyTorch, MLflow); cut out-of-sample RMSE 20–40% against the benchmark under real-time evaluation
- Build implied probability distributions around policy meetings and data releases from option and futures pricing, and judge where the Bank's view differs from what the market has priced
- Produce full predictive distributions rather than point forecasts, used for probability of inflation above the target band in internal briefings
- Read macro headlines as they break and translate them into same-day implications for the forecast; brief the Governing Board on revisions and the macro story behind them around each CPI release; author internal notes and published external research
- Evaluate external data vendors and alternative sources for forecasting value, most recently energy and tourism series; own the models in production and their daily upkeep (PySpark / Spark MLlib on Cloudera DSW, GitLab CI/CD, internal R and Python packages, monitoring dashboards)

Roland Berger

Zurich, Switzerland

External Expert, Data Science

Jun 2022 – Aug 2022

- Engaged as external quantitative expert to build machine and deep learning inflation forecasting models for the client
- Advised consulting teams on preprocessing, feature engineering and model selection for macro time series

University of Zurich

Zurich, Switzerland

Head Teaching Assistant, Prof. Michael Wolf, Department of Economics

Feb 2016 – Jan 2025

- Led tutorials for up to 400 students in Statistics, Financial Econometrics, Time Series Analysis and Advanced Statistics (BA and MSc level); organised and supervised examinations across the lecture portfolio

Akina Partners / Ameropa AG

Zurich & Basel, Switzerland

Internships — Private Equity Sales; Trade Finance

2015 – 2016

- Market and investor research for private equity distribution (Akina); treasury support for physical commodity trading and operations (Ameropa)

RESEARCH

Measuring Economic Outlook in the News

with Franziska Eckert, Linus Kühne, Helge Liebert and Rina Rosenblatt-Wisch

R&R, International Journal of Forecasting

The Hedged Random Forest

with Damian Kozbur and Michael Wolf

R&R, Journal of Financial Econometrics

Forecasting Inflation With the Hedged Random Forest

with Michael Wolf

Empirical Economics (2026)

Improved Inference in Financial Factor Models

with Gianluca De Nard and Michael Wolf

International Review of Economics and Finance (2023)

EDUCATION

University of Zurich

Zurich, Switzerland

PhD in Finance — 5.8/6.0 (Swiss scale, 6.0 = max), *Summa Cum Laude*

Sep 2021 – Jun 2025

- Developed machine/deep learning methods for financial time series forecasting
- Doctoral coursework in asset pricing, derivatives pricing and financial econometrics; completed alongside the SNB role

University of Chicago, Booth School of Business

Chicago, USA

Visiting PhD Student

Mar 2025 – Jun 2025

- Hosted by Prof. Christian B. Hansen

ETH Zurich

Zurich, Switzerland

MSc Statistics, Focus: Machine and Deep Learning — 5.4/6.0, *Magna Cum Laude*

Sep 2017 – Feb 2020

- Thesis on domain generalization of artificial neural networks (Python, PyTorch, TensorFlow, Keras)

University of Zurich

Zurich, Switzerland

BA Banking and Finance — 5.5/6.0, *Summa Cum Laude*

Sep 2014 – Sep 2017

- Thesis based on simulation and estimation work in R and MATLAB

SKILLS

- **Programming:** Python (numpy, pandas, scikit-learn, statsmodels, sktime, PyTorch, LangChain), web scraping (requests, BeautifulSoup, Scrapy, Selenium), R, MATLAB, SQL, Spark / PySpark
- **Market data:** Bloomberg, Refinitiv / LSEG, Datastream, FRED, SNB / BIS data APIs
- **Methods:** time series econometrics, point and density forecasting with out-of-sample evaluation, factor models, covariance estimation and shrinkage, ensemble methods, deep learning, NLP and large language models, option-implied probability distributions
- **Production:** Git and GitLab CI/CD, MLflow, package development and internal distribution (CRAN, PyPI), Cloudera DSW
- **Languages:** fluent English and German; intermediate French and Italian